Functional Neuroanatomy of Threat, Memory, and Regulation

Chapter 2

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Central question. How do the structures and systems of the human nervous system generate threat responses, store and update emotional memory, and support regulation—and what can, and cannot, be inferred from evidence about them in traumatic stress?

Learning objectives

By the end of this chapter, doctoral learners should be able to:

- describe the principal divisions of the nervous system and use anatomical terms of position and section accurately;
- explain the organisation of the cerebral cortex, the hippocampal formation, the amygdala, the insula, the anterior cingulate cortex, the brainstem and the cerebellum at a level sufficient for critical reading of trauma neuroscience;
- evaluate claims that a structure or circuit "causes" post-traumatic stress, distinguishing correlational imaging findings from mechanistic evidence;
- compare the strengths and limits of lesion, animal, pharmacological, electrophysiological and neuroimaging methods;
- apply anatomical reasoning to formulation, psychoeducation and service design without treating biology as destiny;
- identify ethical, cultural and data-governance questions raised by neuroanatomical research on trauma.

Introduction

Trauma neuroscience is often taught as a list of regions with assigned emotions: the amygdala for fear, the hippocampus for memory, the prefrontal cortex for control. That shorthand is memorable and almost always misleading. Each structure participates in many processes; each process recruits many structures; and the same anatomy supports adaptive responses in one context and distressing responses in another. A doctoral-level account therefore has to do two things at once. It must give the reader enough anatomical detail to follow the primary literature, and it must teach a disciplined habit of interpretation that resists converting a group-level correlation into a claim about a person.

This chapter supplies both. Sections 2.1 and 2.2 establish the vocabulary and the large-scale layout of the nervous system. Sections 2.3 to 2.9 examine the structures most often implicated in traumatic stress: the amygdala, hippocampal formation, prefrontal cortex, insula, anterior cingulate cortex, brainstem and cerebellum. Section 2.10 turns from individual structures to large-scale networks. The final sections address measurement, cultural and international variation, professional application, ethics, policy and research design.

Three cautions recur throughout. First, anatomy is not function. A structure's presence in a circuit does not tell us what computation it performs, and a bold spot on a brain image does not identify a psychological process. Second, most human trauma research is observational. It can establish association, sequence and plausibility, but rarely the kind of causal isolation available in controlled animal experiments—and animal models carry their own problems of translation. Third, clinical usefulness is a separate question from statistical significance. A finding may be reliable and still too imprecise to guide an individual decision.

> **A note on terminology.** *Trauma in this chapter means exposure to an event or set of conditions that overwhelms ordinary adjustment and may produce persistent distress. It does not name a brain state, a diagnosis, or a fixed neurological signature. Post-traumatic stress disorder (PTSD) is a clinical syndrome defined by diagnostic criteria; most people exposed to trauma do not develop it, and many who develop it recover. The neurobiology described here concerns processes that can be involved in distress and recovery, not a test for who has suffered.*

2.1 Orientation to the nervous system

Divisions and embryological logic

The nervous system is conventionally divided into the central nervous system (CNS)—the brain and spinal cord—and the peripheral nervous system, which connects the CNS to skin, muscle, organs and sensory receptors. The peripheral system is further divided into somatic and autonomic components; the autonomic system into sympathetic, parasympathetic and enteric divisions. These divisions are useful for organising teaching, but they describe overlapping control systems rather than separate machines. The vagus nerve, for example, carries both ascending visceral information and descending parasympathetic output, and its activity is integrated with brainstem and forebrain circuits.

The adult layout is easier to remember when it is derived from development. The neural tube gives rise to three primary vesicles. The prosencephalon becomes the forebrain, dividing into the telencephalon—cerebral hemispheres, basal ganglia, hippocampus and olfactory structures—and the diencephalon—thalamus, hypothalamus, epithalamus and subthalamus. The mesencephalon becomes the midbrain. The rhombencephalon becomes the hindbrain, dividing into the metencephalon (pons and cerebellum) and the myelencephalon

(medulla). The caudal neural tube becomes the spinal cord. The ventricular system follows the same geometry: lateral ventricles in the hemispheres, the third ventricle in the diencephalon, the cerebral aqueduct in the midbrain, and the fourth ventricle between pons and cerebellum. This developmental logic explains why structures that look widely separated in the adult brain—for example, the amygdala and the hippocampus—share embryological origins in the temporal lobe and remain functionally intertwined.

Planes, position and the problem of the cephalic flexure

Neuroanatomical description uses three standard planes. Coronal sections divide the brain into anterior and posterior parts; sagittal sections divide it into left and right; horizontal (axial) sections divide it into superior and inferior parts. Because the human nervous system bends at the junction of midbrain and diencephalon—the cephalic flexure—the same words change meaning between the forebrain and the hindbrain. *Dorsal* means toward the back in the hindbrain but toward the top of the head in the forebrain. *Rostral* points toward the top of the head in the hindbrain but toward the face in the forebrain. *Anterior* and *posterior*, by contrast, follow the long axis of the body and therefore keep a consistent spatial meaning. Confusion about these terms is not a trivial matter of vocabulary: it produces real errors in reading lesion and imaging papers, particularly when coordinates are reported in different conventions.

Grey matter, white matter and the meaning of "connection"

Grey matter contains neuronal cell bodies, dendrites, synaptic neuropil and glia. White matter contains myelinated axons bundled into tracts. Cortical grey matter is organised in layers, and the relative prominence of different layers varies across the cortex, producing the cytoarchitectonic maps associated with Brodmann and later quantitative atlases. Subcortical grey matter forms nuclei—clusters of cell bodies with relatively distinct inputs and outputs—such as the thalamus, basal ganglia, amygdala and brainstem nuclei. White-matter tracts connect these regions. The corpus callosum carries hundreds of millions of axons between hemispheres; the internal capsule carries ascending and descending fibres past the basal ganglia; the fornix connects hippocampus to diencephalon and other targets.

The distinction matters for interpretation. A structural finding in grey matter may reflect cell bodies, dendrites, synapses or vasculature; a diffusion finding in white matter may reflect myelin, axon calibre, fibre

packing or extracellular water. Neither is a direct measure of "connectivity" in the functional sense. When a paper reports reduced white-matter integrity in trauma-exposed groups, the appropriate conclusion concerns a statistical property of a diffusion metric in that sample—not a severed pathway or a broken relationship.

The brainstem, cranial nerves and ascending systems

The brainstem comprises the midbrain, pons and medulla. It contains cranial nerve nuclei serving sensation and movement of the head and neck, ascending and descending long tracts, and the reticular formation—a distributed network of nuclei involved in arousal, sleep–wake regulation, autonomic control, pain modulation and motor tone. Several nuclei project diffusely to the forebrain and are therefore described as neuromodulatory: the locus coeruleus (noradrenaline), the raphe nuclei (serotonin), the ventral tegmental area and substantia nigra (dopamine), and the basal forebrain cholinergic system. Because these nuclei influence large territories, their involvement in stress responses can produce widespread effects that are easy to mistake for regionally specific pathology.

The brainstem also contains the periaqueductal grey (PAG), discussed in Section 2.8, and the parabrachial nucleus, which relays visceral and nociceptive information to the amygdala, insula and hypothalamus. These pathways provide an anatomical basis for the bodily character of threat: the sense of pressure in the chest, the sudden nausea, the impulse to freeze. They do not by themselves explain the meaning a person assigns to those sensations.

Vasculature, meninges and the limits of inference from lesions

The brain is supplied by the anterior and posterior circulations and drained by venous sinuses; the meninges (dura, arachnoid and pia) enclose the CNS and the cerebrospinal fluid. Vascular territory matters clinically because infarcts produce relatively stereotyped lesion patterns, and historically these patterns supplied much of what was known about structure–function relationships. The classic example—patient H.M.'s bilateral medial temporal resection and subsequent anterograde amnesia—established the importance of the medial temporal lobe for declarative memory (Scoville & Milner, 1957). But lesions are rarely pure. They remove fibres passing through as well as cell bodies, they disrupt blood supply to adjacent tissue, and they trigger plasticity over time. A lesion study can demonstrate that a region

is necessary for a function in a particular patient; it cannot by itself specify the mechanism.

2.2 Cerebral cortex and subcortex

The cortex as a continuous sheet

The cerebral cortex is a thin, highly folded sheet of grey matter, typically three to five millimetres thick, containing neuronal cell bodies, dendrites, synapses, glia and blood vessels. Folding increases surface area within the cranial cavity: ridges are gyri, valleys are sulci, and especially deep sulci are fissures. The major lobes—frontal, parietal, temporal and occipital—are defined largely by surface landmarks. The central sulcus separates frontal from parietal cortex on the lateral surface; the lateral (Sylvian) fissure separates the temporal lobe from the frontal and parietal lobes above; the parieto-occipital sulcus marks the medial boundary between parietal and occipital cortex.

These boundaries are conventions, not functional compartments. The cortical sheet is continuous, and the same cytoarchitectonic field can extend across a named boundary. The insula, for example, is buried beneath the opercular cortex of the frontal, parietal and temporal lobes; it is not an island but a continuously connected region hidden by the disproportionate growth of surrounding cortex. Contemporary accounts therefore prefer to describe gradients and networks rather than treating lobes as functional units.

Cytoarchitecture and cortical hierarchies

Cortical areas differ in the relative thickness of their six layers. Primary sensory cortices have a prominent granular layer IV, reflecting dense thalamic input. Primary motor cortex has a prominent layer V, reflecting output to brainstem and spinal cord. Association cortices show less distinct lamination and extensive recurrent and long-range connectivity. This architecture supports a rough hierarchy: unimodal sensory cortex, then multimodal association cortex, then paralimbic and limbic regions that integrate sensory information with internal state, memory and value.

Hierarchy is not the same as control. The phrase "top-down control" is often used as though higher areas issue commands to lower ones. In practice, cortical and subcortical systems interact reciprocally, and the direction of influence depends on the task and the state of the organism. The prefrontal cortex can bias processing in sensory and limbic regions,

but it also receives dense ascending input that shapes what it can compute. Under acute stress, ascending noradrenergic and dopaminergic signals can rapidly alter prefrontal function—sometimes improving simple vigilance while impairing complex working memory.

Subcortical structures

The thalamus is the principal relay for sensory and motor information to the cortex, and it participates in cortico-thalamo-cortical loops that support attention and cognition. The hypothalamus integrates endocrine, autonomic and behavioural responses through its control of the pituitary and its projections to brainstem nuclei. The basal ganglia—caudate, putamen, globus pallidus, subthalamic nucleus and substantia nigra—support action selection, habit learning and reward-based learning. The amygdala and hippocampal formation, treated in the following sections, are anatomically adjacent but functionally distinct.

The basal ganglia deserve more attention in trauma teaching than they usually receive. Avoidance, habitual safety behaviour and reward learning are central to how traumatic stress persists, and these processes depend on corticostriatal circuits. A narrow focus on the amygdala as the "fear centre" can obscure the role of striatal learning in the gradual construction of a narrowed life.

Clinical illustration

Consider a patient who avoids driving after a collision. An amygdala-centred explanation would emphasise conditioned fear to the vehicle. A fuller formulation would also consider hippocampal contextual memory (the specific junction where the collision occurred), prefrontal appraisal (the belief that driving is now inherently dangerous), insular interoception (the nausea and palpitations that precede avoidance), basal ganglia habit (the practised route of avoidance), and brainstem arousal (sleep disruption and exaggerated startle). Each level suggests a different intervention: exposure to the avoided context, cognitive work on appraisal, interoceptive awareness, sleep treatment, or graded return to driving. The anatomy is useful precisely because it generates multiple hypotheses rather than one.

2.3 Amygdala organisation

Structure

The amygdala is an almond-shaped group of nuclei in the medial temporal lobe, anterior to the hippocampus. Its principal subdivisions include the lateral, basal and accessory basal nuclei (often grouped as the basolateral complex), the central nucleus, the medial nucleus and the cortical nuclei, together with the intercalated cell masses that sit between the basolateral complex and the central nucleus. The bed nucleus of the stria terminalis (BNST) is a closely related structure in the extended amygdala, continuous with the central nucleus and involved in sustained anxiety rather than brief phasic fear.

The flow of information is roughly serial and parallel. Sensory information reaches the lateral and basal nuclei from thalamus and cortex. These nuclei project to the central nucleus, directly and through the intercalated cells. The central nucleus, in turn, projects to the hypothalamus, brainstem and PAG, producing autonomic, endocrine and behavioural components of threat responses. The intercalated cells are inhibitory and are thought to gate information from basolateral to central nuclei; they have been implicated in the expression of extinction memories (Milad & Quirk, 2012). Extinction does not erase the original association. It establishes a new, context-dependent inhibitory memory that competes with the original, which is why fear can return after a change of context, the passage of time or a new stressor. Reconsolidation research suggests that a memory can become temporarily labile after retrieval and may be modified during restabilisation (Nader et al., 2000), but the conditions under which this can be used therapeutically in humans remain contested.

Function

The amygdala is not a fear module. It responds to novelty, salience, uncertainty, reward prediction and social signals, and its role in threat learning is best understood as part of a broader system for detecting and evaluating biologically significant events. In Pavlovian fear conditioning, a neutral conditioned stimulus is paired with an aversive unconditioned stimulus; over trials, the conditioned stimulus comes to elicit defensive responses. The lateral amygdala is a principal site of plasticity for this association, while the central nucleus organises the output (LeDoux, 2000; Maren, 2001). Lesions of the amygdala impair acquisition and expression of conditioned fear but do not abolish all defensive behaviour, which can also be organised by brainstem and hypothalamic circuits.

Human neuroimaging consistently shows amygdala responses to threat-related stimuli, including faces, scenes and trauma reminders. In PTSD, several meta-analyses report exaggerated amygdala reactivity to trauma-related or threat-related cues compared with trauma-exposed controls, together with reduced medial prefrontal engagement. These are group-level differences with substantial overlap between groups; they cannot be used to diagnose an individual, and their direction can vary with task, symptom profile and dissociation.

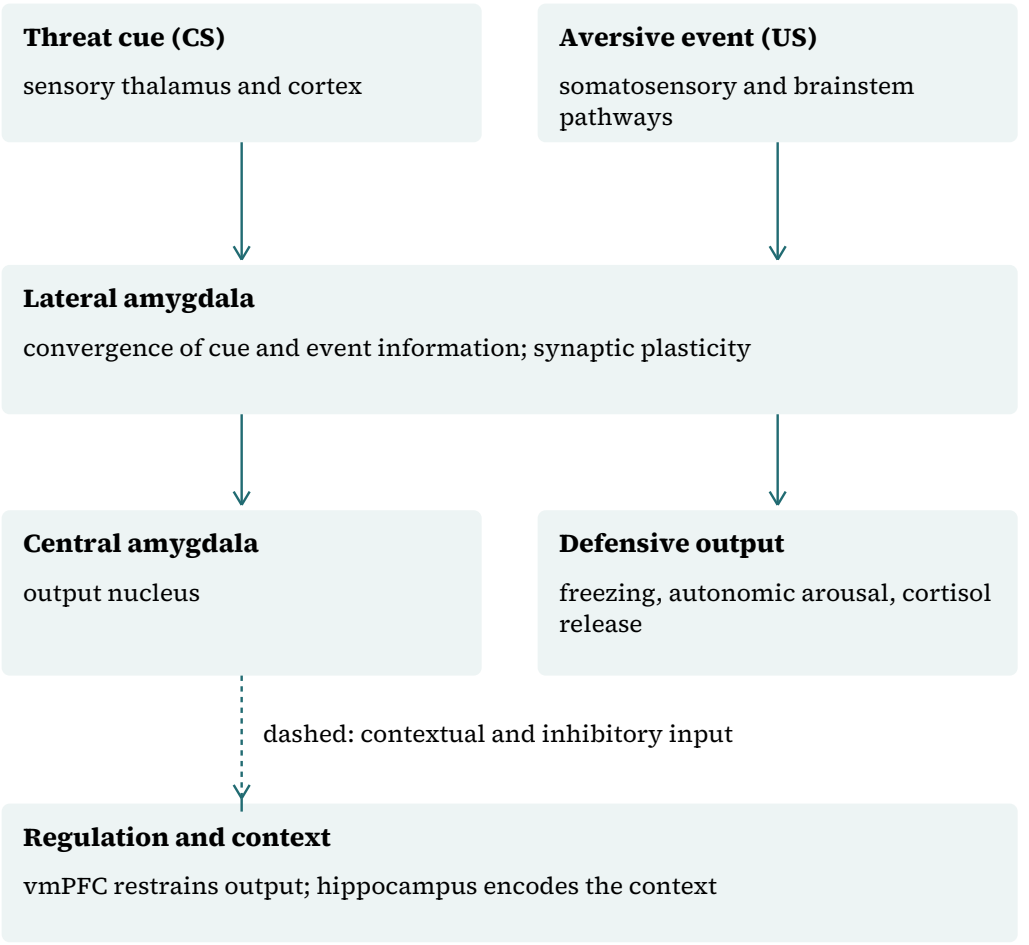
Avoidance and behavioural engagement

Recent work has broadened the amygdala's role from fear conditioning to decision-making about engagement with the environment. In rodent models, passive freezing and active avoidance depend on partially distinct pathways: lateral amygdala to central amygdala to PAG for freezing, and lateral amygdala to basal amygdala to nucleus accumbens for active avoidance, with infralimbic cortex contributing to the latter. These findings matter clinically because avoidance is not a single behaviour. A patient who freezes, a patient who flees, and a patient who quietly reorganises life to avoid reminders may be expressing different circuits and may need different therapeutic emphases.

Caveats

Most detailed microcircuit evidence comes from rodents. Homologies between rodent and human prefrontal cortex are debated, and the translation of optogenetic findings to human PTSD is indirect. Human imaging measures are sensitive to movement, physiological noise, task design and analysis choices. The amygdala is a small, heterogeneous structure; standard voxel sizes blur its subdivisions, so a "basolateral" finding in a typical fMRI study is often an inference rather than a measurement. These limits do not make the findings worthless, but they should temper claims that a specific nucleus has been shown to cause a specific symptom in a patient.

Figure 2.1. A simplified threat-learning circuit



CS, conditioned stimulus; US, unconditioned stimulus; vmPFC, ventromedial prefrontal cortex. Connections are simplified: the amygdala is treated as a functional unit and many parallel routes are omitted.

2.4 The hippocampal formation

Structure

The hippocampal formation lies in the medial temporal lobe and includes the dentate gyrus, the hippocampus proper (cornu ammonis fields CA1, CA2, CA3 and the hilus), the subiculum and the adjacent presubiculum and parasubiculum, with the entorhinal cortex as its principal cortical interface. The dentate gyrus receives input from entorhinal layer II via the perforant path and projects to CA3 via mossy fibres; CA3 projects to CA1 via Schaffer collaterals; CA1 projects to the subiculum and back to entorhinal cortex. This trisynaptic organisation is

the anatomical basis of a widely studied model of learning and memory, although the system also contains extensive parallel and recurrent connections that the simple circuit diagram omits.

The hippocampus is not a single homogeneous structure. Its dorsal (in humans, posterior) and ventral (anterior) portions differ in connectivity and function, with the posterior hippocampus more strongly associated with spatial and episodic memory and the anterior portion more strongly connected to amygdala and affective processing. This gradient is relevant to trauma research, where "hippocampal volume" is often treated as a single variable.

Function

The hippocampal formation is essential for the formation of declarative memory—memory that can be consciously recalled, including episodic memory for events and semantic memory for facts—and for spatial navigation and contextual processing. It supports relational memory: the ability to bind elements of an experience into a coherent representation and to use that representation flexibly (Squire, 2004). It is also involved in imagining future events, which is why hippocampal damage can impair both remembering and prospecting.

In threat learning, the hippocampus provides the contextual frame. A conditioned cue may signal danger only in a particular place, time or internal state; the hippocampus is central to representing that context and to the renewal of fear when a context changes. This has clinical implications: a patient may feel relatively safe at home and overwhelmed in a supermarket, not because the supermarket is inherently more dangerous but because contextual processing has been altered by trauma.

Findings in PTSD and their interpretation

Reduced hippocampal volume is one of the most replicated structural findings in PTSD, reported across trauma types and supported by large imaging consortia. The interpretation, however, is unresolved. Volume reduction may precede trauma and confer vulnerability; it may follow trauma through glucocorticoid-related dendritic remodelling; it may reflect comorbid depression, alcohol use, sleep disturbance or medication; and it may be partly explained by image-processing choices. Longitudinal studies are difficult and relatively few. The appropriate conclusion is that smaller hippocampal volume is associated with PTSD in many samples, that the causal direction is uncertain, and that hippocampal volume is not a diagnostic test.

Animal work provides plausible mechanisms. Chronic stress and glucocorticoid exposure can reduce dendritic arborisation in CA3, suppress neurogenesis in the dentate gyrus and alter long-term potentiation in CA1 (McEwen, 1998). These findings are important, but they come from controlled paradigms with species-specific stress exposures. They establish that stress can alter hippocampal structure and function under particular conditions; they do not establish that a given patient's scan reflects a particular life event.

Clinical illustration

A patient reports fragmented recall of a road collision but vivid, intrusive images of a specific detail—the sound of braking. A hippocampal account would emphasise impaired contextual binding: the fragment is not integrated into a coherent episode with a beginning, middle and end. A therapeutic implication is that rebuilding a coherent narrative, in a safe and paced way, may reduce the fragment's intrusive quality. This is a hypothesis, not a proven mechanism, and it should be offered as part of a broader formulation rather than as a neuroanatomical certainty.

2.5 Prefrontal cortical systems

Structure

The prefrontal cortex occupies the anterior portion of the frontal lobe. Major subdivisions include the dorsolateral prefrontal cortex (dlPFC), important for working memory, planning and cognitive flexibility; the ventrolateral prefrontal cortex, involved in response selection and inhibition; the ventromedial prefrontal cortex (vmPFC), including medial orbital and subgenual regions, involved in value-based decision-making and emotion regulation; and the anterior cingulate cortex (Section 2.7). These regions are densely interconnected with thalamus, basal ganglia, amygdala, hippocampus and brainstem neuromodulatory systems.

Function and stress effects

The prefrontal cortex supports the maintenance and manipulation of information in working memory, the inhibition of prepotent responses, the flexible updating of rules, and the regulation of emotion through reappraisal and related strategies (Ochsner & Gross, 2005). It is also among the most stress-sensitive cortical regions. Work by Arnsten and

colleagues showed that even mild, uncontrollable stress can rapidly impair prefrontal working memory in animals and humans, through high levels of noradrenaline and dopamine acting on alpha-1 and D1 receptors. These signals engage intracellular cascades—calcium–cAMP signalling and protein kinase C—that reduce prefrontal firing and weaken network connectivity. With chronic stress or glucocorticoid exposure, dendritic spines on prefrontal pyramidal cells retract, while dendrites in the amygdala expand. The same neurochemical events that impair prefrontal function can strengthen amygdala-dependent emotional learning, shifting control toward more reflexive responses.

This is one of the clearest mechanistic accounts in the field, and it is worth stating precisely. It does not show that stress damages the prefrontal cortex permanently, nor that trauma survivors have a fixed prefrontal deficit. It shows that under conditions of uncontrollable stress, prefrontal networks become less effective and subcortical defensive systems become more influential—an adaptive shift in the short term, potentially costly when sustained.

Imaging findings in PTSD

Functional imaging in PTSD frequently reports reduced activation of the vmPFC and rostral anterior cingulate during symptom provocation or extinction recall, and reduced functional connectivity between vmPFC and amygdala. These findings are consistent with the idea that extinction memory retrieval is impaired, but they are not uniform. Some studies report increased prefrontal activation during sustained attention, possibly reflecting compensatory effort, and the pattern varies with dissociation, comorbidity and task demands.

The most defensible reading is that PTSD is associated with altered prefrontal regulation of limbic and autonomic systems in many—not all—samples, and that the direction and magnitude of the difference depend on what the person is doing and how the task is designed. A single scan cannot establish that a person's prefrontal cortex is "underactive" in daily life.

Implications for intervention

Prefrontal accounts support interventions that strengthen regulatory capacity: sleep restoration, reducing unnecessary cognitive load, practising reappraisal, and rebuilding predictable routines. They also caution against overloading survivors with complex cognitive demands during acute stress. But "strengthening the prefrontal cortex" is not a clinical prescription. The evidence for generalised cognitive training

improving trauma symptoms is weak, and the most effective treatments—trauma-focused psychotherapy and, for some, medication—work through multiple mechanisms that are not reducible to prefrontal activation.

2.6 Insula and interoception

Structure

The insula is a region of cortex buried within the lateral sulcus, bounded by the frontal, parietal and temporal opercula. It is conventionally divided into posterior, mid and anterior zones. The posterior insula receives visceral and somatosensory input and contains a primary interoceptive representation; the mid insula integrates this information with sensory and affective signals; the anterior insula is connected with anterior cingulate, prefrontal, amygdala and reward-related structures. The insula is also a core node of the salience network (Section 2.10).

Interoception

Interoception refers to the sensing and processing of signals from within the body, including cardiac, respiratory, gastrointestinal and visceral information. It is not a single sense. It includes the detection of internal signals (interoceptive accuracy), the subjective belief about one's sensitivity (interoceptive sensibility), and the correspondence between them (interoceptive awareness). These components are measured differently and do not always agree.

Craig's account proposes a posterior-to-anterior progression in which primary interoceptive representations are re-represented in the anterior insula, providing a basis for subjective feelings and emotional awareness. This is an influential model, but its stronger claims—particularly the proposal that the anterior insula is a neural correlate of consciousness—remain contested. For trauma teaching, the more defensible points are that interoceptive signals contribute to emotional experience, that the insula participates in detecting and integrating them, and that individuals differ markedly in interoceptive accuracy and in how they interpret bodily sensations.

Relevance to traumatic stress

Trauma-related distress is frequently bodily. Panic-like sensations, nausea, tightness, numbness and depersonalisation involve altered

interoceptive processing. Some studies report altered interoceptive accuracy in PTSD and in related conditions, but findings are inconsistent and depend on the task and the comparison group. The clinical value of the interoceptive framework is that it provides a non-mystical language for bodily symptoms and suggests graded, tolerable exposure to internal sensations rather than avoidance.

Dissociation complicates the picture. In the dissociative subtype of PTSD, depersonalisation and derealisation involve a sense of detachment from the body or surroundings. Imaging studies report a pattern opposite to classic re-experiencing: increased vmPFC and dorsal anterior midcingulate activity, reduced amygdala and insula activity, and blunted or reduced heart-rate responses to trauma cues. This does not mean that dissociation is "shutting down" the brain; it suggests a different regulatory configuration, with top-down overmodulation of limbic and autonomic responses.

2.7 Anterior cingulate cortex

Structure and subdivisions

The cingulate cortex is a belt of cortex encircling the corpus callosum on the medial surface of the hemispheres. It is divided into anterior, mid and posterior regions, with further subdivisions based on architecture, connectivity and function. The anterior cingulate cortex (ACC) includes rostral and ventral portions associated with affective processing and autonomic regulation. The midcingulate cortex (MCC), often described as dorsal ACC, includes an anterior portion (aMCC) that has become central to integrative accounts of negative affect, pain and cognitive control.

Integration rather than segregation

A long-standing model assigned emotion to rostral ACC and cognition to dorsal MCC. Shackman and colleagues reviewed physiological, anatomical and functional evidence and concluded that this segregation is difficult to sustain. Negative affect, pain and cognitive control all engage an overlapping region of the dorsal cingulate—the aMCC—which appears to function as a hub linking information about reinforcers to motor centres responsible for expressing affect and executing goal-directed behaviour. On this account, aMCC implements a domain-general process that uses punishment-related information to adapt behaviour.

This is a useful corrective to the habit of labelling ACC activity as "conflict monitoring" or "pain" or "emotion" depending on the study. The same region can be active in many tasks because it is doing something common to them—evaluating the significance of events for action—rather than because the tasks share a hidden emotional content.

Relevance to trauma

ACC and MCC findings in PTSD include altered activation during symptom provocation, altered connectivity with amygdala and insula, and associations between rostral ACC activity and extinction recall. These findings are embedded in the larger prefrontal-limbic literature and should not be read in isolation (Pitman et al., 2012). A particularly important caution concerns pain. Trauma survivors frequently report pain, and the overlap between pain and negative affect in aMCC means that a trauma-informed explanation should not reduce pain to "psychological" causation, nor treat it as unrelated to stress. Both dimensions require assessment.

2.8 Brainstem and periaqueductal grey

The brainstem as an integrative centre

The brainstem is not a passive conduit. It contains the cranial nerve nuclei that serve vision, hearing, taste, facial sensation and movement, swallowing, and parasympathetic control of the heart and viscera. It contains the reticular formation, with ascending systems that regulate cortical arousal and descending systems that modulate pain and autonomic output. It contains the locus coeruleus, the principal source of noradrenaline in the forebrain, and the raphe nuclei, the principal source of serotonin. It also contains the PAG.

The periaqueductal grey

The PAG surrounds the cerebral aqueduct in the midbrain. It is organised into columns with different connectivity and different defensive functions. The dorsolateral and dorsomedial columns have been associated with active defensive responses such as flight and fight; the lateral column with freezing and vigilance; the ventrolateral column with passive coping, quiescence and immobilisation. These associations come largely from animal work and are not rigid; the same column can support different behaviours depending on context and on inputs from amygdala, hypothalamus and prefrontal cortex.

The PAG is a plausible node in the circuitry of trauma-related freezing, shutdown and dissociation. Fenster and colleagues note increased functional connectivity of the ventrolateral PAG with regions linked to passive responses to threat in the dissociative subtype of PTSD compared with classic PTSD. This is an interesting convergence between animal defensive circuitry and human symptom patterns, but it remains a correlational finding in a modest literature.

Arousal, sleep and the body

Brainstem nuclei regulate sleep–wake transitions, REM sleep, breathing and heart rate. Trauma-related sleep disturbance—fragmented sleep, nightmares, hyperarousal at night—is therefore not merely a cortical phenomenon. Noradrenergic activity is reduced during REM sleep under ordinary conditions, and alterations in this system may contribute to the intrusive, emotionally intense quality of traumatic dreams. Sleep disruption, in turn, affects prefrontal function, emotional reactivity and pain sensitivity, creating feedback loops that maintain distress.

Clinical implications

Brainstem reasoning supports attention to basic physiology: breathing, sleep, heart rate, medication effects, thyroid disease, anaemia and substance withdrawal can all mimic or amplify trauma symptoms. It also supports caution about purely psychological formulations of symptoms such as collapse, numbness or inability to move during threat. These responses may reflect conserved defensive systems rather than weakness or unwillingness.

2.9 Cerebellar contributions

Structure

The cerebellum lies behind the brainstem and beneath the occipital lobes. It has a highly folded cortex, deep nuclei (dentate, interposed and fastigial), and extensive connections with the cerebral cortex via the pons and thalamus, and with the spinal cord via the brainstem. Its traditional association is with motor coordination, balance and motor learning. That association is correct but incomplete.

Functional topography

Stoodley and Schmahmann's activation-likelihood meta-analysis of neuroimaging studies found evidence for functional topography within the human cerebellum. Sensorimotor tasks activated the anterior lobe, particularly lobule V and adjacent lobule VI, with additional foci in lobule VIII. Language and verbal working memory engaged lobule VI and Crus I; spatial tasks engaged lobule VI; executive tasks engaged lobules VI, Crus I and VIIB; emotional processing engaged lobules VI, Crus I and medial VII. The posterior lobe was consistently involved in higher-level tasks, consistent with the proposal that distinct cerebellar regions participate in cerebro-cerebellar loops with motor, cognitive and affective cortical areas.

Why this matters for trauma

The cerebellum is not a primary trauma structure, and claims that it "stores trauma" should be treated with scepticism. Its relevance is indirect but real. Cerebellar loops contribute to timing, prediction, error-based learning and the regulation of autonomic and emotional responses. Cerebellar dysfunction can produce symptoms—dizziness, tremor, unsteadiness, abnormal eye movements—that overlap with dissociation and anxiety and that require neurological assessment. Trauma-informed clinicians should be able to recognise when a presentation needs medical review rather than assuming a psychological cause.

The cerebellum also illustrates a broader lesson. A structure can be meaningfully involved in a process without being the site of a disorder. The question is always functional: what computation does this region contribute, under what conditions, and how would we know?

2.10 Large-scale brain networks

From regions to networks

The human brain is organised into large-scale intrinsic connectivity networks—sets of regions that show correlated activity at rest and co-activate across tasks. Three networks have been particularly influential in psychiatry. The central executive network (CEN), anchored in dorsolateral prefrontal and posterior parietal cortex, supports working memory and goal-directed control. The default mode network (DMN), anchored in medial prefrontal cortex, posterior cingulate and angular gyrus, supports self-referential thought, autobiographical memory and social cognition (Menon, 2023). The salience network (SN),

anchored in anterior insula and dorsal anterior cingulate, detects biologically significant events and coordinates switching between CEN and DMN.

Menon's triple network model proposes that many psychiatric conditions involve aberrant salience mapping and impaired access, engagement or disengagement of these networks. In PTSD, studies report altered within-network and between-network connectivity, particularly involving DMN, SN and prefrontal-limbic circuits. These findings are promising but heterogeneous. Connectivity estimates depend on preprocessing, motion correction, atlas choice and analysis strategy; test–retest reliability varies across regions; and group differences are often small relative to individual variation.

What network findings can and cannot establish

A network finding can show that regions covary differently between groups. It cannot show that a network is "overactive" in a person, that a specific thought caused a specific connectivity pattern, or that altering connectivity will relieve symptoms. Network language is nonetheless valuable because it moves the field away from single-region localisation and towards a systems view in which threat detection, memory, interoception and regulation are distributed.

For clinical teaching, the most useful implication is that trauma-related distress is not produced by one faulty node. It is produced by a system that has learned, in a particular body and environment, to prioritise threat. That formulation opens multiple points of intervention—sleep, safety, relationships, meaning, avoidance, physiological regulation—without pretending that any one of them is the whole answer.

Synthesis: from anatomy to formulation

Table 2.1 summarises the structures discussed in this chapter, the processes in which they are most plausibly involved, and the limits that should accompany any clinical or research claim.

Table 2.1. Structures, plausible processes and interpretive limits

Structure or system	Plausible processes	Interpretive limits
Amygdala	Threat detection, Pavlovian fear learning, salience, avoidance	Small and heterogeneous; rodent microcircuits do not map directly onto human symptoms
Hippocampal formation	Declarative and contextual memory, relational binding, spatial navigation	Volume findings are associational; direction and specificity uncertain
Prefrontal cortex	Working memory, inhibition, reappraisal, extinction recall	Acute stress effects are state-dependent; chronic changes are not fixed deficits
Insula	Interoception, salience, emotional awareness	Interoceptive tasks measure different constructs; strong awareness claims are contested
Anterior cingulate	Integration of negative affect, pain and cognitive control	Overlap across domains means activation alone cannot identify a process
Brainstem and PAG	Arousal, autonomic control, defensive behaviour, sleep	Animal-derived defensive columns are not rigid human subtypes
Cerebellum	Timing, prediction, motor and cognitive loops	Indirect relevance to trauma; not a trauma store
Large-scale networks	Distributed coordination of cognition, salience and self-reference	Connectivity depends on analysis choices; group differences are not diagnoses

Table 2.2. Levels of analysis and what each can support

Level	Typical methods	What it can support	What it cannot establish alone
Molecular and cellular	Assays, cell culture, animal models	Mechanisms under controlled conditions	Direct translation to a person's experience
Circuit and systems	Lesions, electrophysiology, optogenetics, imaging	Necessary or sufficient involvement in a task	A diagnosis or a causal life history
Behaviour and function	Interviews, questionnaires, task performance	Functional impact and meaning	Underlying mechanism
Social and implementation	Qualitative, mixed-methods, policy evaluation	Acceptability, access, equity, feasibility	Biological mechanism

Measurement and interpretation

Anatomical claims are only as good as the measurements behind them. Three distinctions are essential.

First, **reliability is not validity**. A measure can be highly reproducible and still not index the construct it is said to index. Amygdala reactivity on fMRI can be reliable across sessions and still reflect arousal, attention, novelty or movement as much as fear.

Second, **group differences are not individual diagnoses**. Most trauma neuroscience compares group means. The distributions overlap substantially, and a threshold that separates groups in a study is rarely suitable for clinical use.

Third, **statistical significance is not clinical utility**. A small association in a large sample may be precise and still useless for deciding what to do for a patient. Conversely, a clinically important effect may fail to reach significance in an underpowered study.

These distinctions should be applied consistently, including to findings that support a favourite theory. The field has a history of overinterpreting each new technique—phrenology, lesion localisation, early PET, candidate genes, and now machine-learning prediction. Each technique added something real; each was also asked to carry more meaning than it could bear.

Cultural and international perspectives

Anatomical knowledge is universal in the sense that human brains share a basic plan. The interpretation of trauma is not. Culture shapes which events are disclosed, which symptoms are salient, how distress is expressed in the body, who is consulted, and what recovery is expected to look like. In many settings, stigma, gender norms, legal status, migration, poverty and the availability of services influence whether a person presents at all.

Cross-cultural research also raises measurement questions. Instruments developed in one language may not be equivalent in another; translation is not validation. Biological measures may be affected by nutrition, infection, medication, sleep and chronic stress in ways that differ across settings. Findings from high-income countries should not be assumed to generalise, and multinational studies should involve local investigators, test measurement equivalence and report constraints honestly.

For the Nigerian and wider African contexts in which this programme is taught, these issues are not peripheral. The evidence base for many trauma interventions was developed elsewhere, and implementation must reckon with workforce shortages, out-of-pocket costs, traditional and religious healing systems, and the role of family and community in recovery. A neurobiological account that ignores these realities will be technically correct and practically useless.

Professional practice

Neuroanatomical knowledge should improve explanation, assessment and referral—not expand the scope of practice beyond qualification.

- **Explain without reducing.** Describe what is known about threat responses in plain language, and state clearly what is not known. Avoid phrases such as "your amygdala is damaged" or "your brain is stuck".
- **Assess before interpreting.** Rule out medical, neurological, pharmacological and substance-related causes of symptoms that resemble trauma responses.
- **Respect the limits of tests.** Do not order imaging, genetic testing or biomarkers for the purpose of "proving" trauma. There is no established clinical indication for these tests in routine PTSD assessment.

- **Use anatomy to generate hypotheses, not conclusions.** A formulation should integrate biological, psychological, social and cultural information and should remain revisable.
- **Protect data.** Brain images, genetic information and clinical records are sensitive. Collect only what is needed, explain retention and sharing, and comply with local law.

Ethics and data governance

Trauma neuroscience carries specific ethical risks. Biological language can stigmatise, imply permanence or shift responsibility from circumstances to individuals. Research can re-traumatise participants, generate incidental findings, or create expectations of clinical benefit that cannot be met. Genetic and neuroimaging data can be re-identifiable and may reveal information about relatives who did not consent.

Good practice includes proportionate collection, secure storage, clear limits on secondary use, accessible explanations, and a plan for incidental findings. Community involvement in design and interpretation is not a formality; it improves the questions asked and the likelihood that findings are used responsibly. Where research is conducted in under-resourced settings, partnerships should build local capacity rather than extract data.

Policy implications

Policy should convert evidence into proportionate, reviewable action. A responsible policy states the problem, population, mechanism, resources, accountable body, implementation plan, indicators and review date. Neuroscience can support investment in prevention, mental healthcare, sleep, occupational safety, maternal and child health, disaster response and research infrastructure. It should not be used to justify surveillance, exclusion, compulsory testing or discrimination when clinical utility and fairness have not been demonstrated.

Policy also has to reckon with uncertainty. A government that waits for definitive neuroscience before acting will fail people; a government that treats preliminary findings as settled will waste resources and may cause harm. The practical task is to act on the best available evidence while building in evaluation and review.

Composite case study

A regional disaster affected several communities after severe flooding and infrastructure failure. A multidisciplinary team observed sleep disruption, heightened startle, grief, pain, displacement, interrupted medication and reduced access to care. The team considered amygdala-dependent threat learning, hippocampal contextual processing, insular interoception, prefrontal regulation and brainstem arousal, but avoided assuming that every response represented a disorder. It created a stepped pathway combining safety, physical healthcare, practical assistance, family contact, psychological first aid, screening only where referral was available, and specialist treatment for persistent impairment. Researchers proposed optional longitudinal measurement, with separate consent, secure data handling, community oversight and feedback of aggregate findings.

The case illustrates a general point. Neurobiological knowledge gains practical value when it supports humane decisions and remains connected to social conditions. The team did not need to know each person's amygdala activity to design a better pathway. It needed to know enough about threat, sleep, memory and regulation to ask the right questions and to avoid harm.

Local composite case study

In a fictional Nigerian teaching hospital, an adult attended after a road traffic collision and later reported nightmares, bodily alarm, poor concentration, pain and fear of travelling. The clinician reviewed injury, medication, sleep, substance use, prior health, current safety, social support and functional change. The formulation used amygdala-dependent threat learning, hippocampal contextual memory, interoceptive signals and prefrontal regulation as possible explanatory concepts, not as proof of diagnosis. The team provided plain-language education, coordinated medical and psychological review, agreed measurable goals, and arranged follow-up.

A student research group considered studying the pathway but first examined language validity, transport barriers, phone access, follow-up loss, and the danger of collecting biological samples without clear clinical or research value. Their caution is part of the science, not an obstacle to it.

Worked example

A learner must decide whether a reported association between hippocampal volume and trauma symptoms supports a causal claim.

State the exposure, outcome, population and timing.

Identify the design and whether temporal order is established.

Examine measurement reliability and validity, including scanner and segmentation choices.

List plausible confounders: age, sex, education, alcohol use, depression, sleep, medication, medical illness and socioeconomic adversity.

Inspect effect size and uncertainty, not only the p value.

Ask whether the finding replicated and whether the comparison group was appropriate.

Restate the conclusion at the level the evidence supports.

A defensible conclusion will usually describe an association in a defined population and specify what design would strengthen causal inference. It will not say that trauma shrank the hippocampus in a particular patient.

Review and discussion questions

Why is it misleading to describe the amygdala as the brain's "fear centre"? What would be a more accurate formulation?

Explain the difference between grey-matter volume, white-matter integrity and functional connectivity. What does each measure, and what does it not measure?

How does the cephalic flexure affect the use of dorsal, ventral, rostral and caudal in neuroanatomy?

What evidence would be required to show that a brain region causes a PTSD symptom rather than merely correlating with it?

Why do group-level imaging findings have limited value for diagnosing an individual?

How should a clinician respond to a patient who asks whether a scan can prove that they have trauma?

What are the ethical risks of researching trauma neurobiology in a low-resource setting, and how can they be reduced?

How might cultural context change the interpretation of bodily symptoms that are assumed to be interoceptive?

Research assignments

- Design a study that could distinguish pre-existing hippocampal differences from trauma-related change. Specify sampling, timing, comparison group and analysis.
- Critically appraise one meta-analysis of amygdala or prefrontal findings in PTSD. Report effect sizes, heterogeneity, publication bias and the limits of the included designs.
- Write a protocol for a cross-cultural validation of an interoception measure, including translation, cognitive interviewing and measurement invariance.
- Propose a study of defensive responses in humans that respects the ethical limits on inducing threat. Explain what the study could and could not conclude.

Chapter summary

This chapter described the anatomical systems most often implicated in threat, memory and regulation. The amygdala supports threat learning and salience, the hippocampal formation supports declarative and contextual memory, the prefrontal cortex supports working memory and regulation, the insula supports interoception, the anterior cingulate integrates affective and cognitive demands, the brainstem organises arousal and defensive behaviour, and the cerebellum contributes to timing and prediction. Large-scale networks coordinate these systems.

The central lesson is not anatomical memorisation but calibrated inference. Each structure participates in many processes; each method answers a narrow question; each finding is embedded in a design, a sample and an analysis. Doctoral-level practice requires the discipline to state what the evidence supports, to preserve what remains uncertain, and to design the next question so that uncertainty is reduced rather than disguised.

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